Mississippi Valley State University

CS 341 — Discrete Structures

Lecture 23: When Direct Proof Stalls

Proof by Contrapositive

Chartrand & Zhang, Section 3.3

Name: _____ Date: _____

Where we are

Wednesday's method works when the hypothesis gives you something to compute with. Sometimes it does not — and then the same statement, written backwards and negated, becomes easy. You have had the tool since Week 2.

By the end of today you should be able to

1. Recognise when a direct proof will stall.
2. Write a proof by contrapositive with the correct opening.
3. Spot a proof that proves the wrong implication.

1. The method

Proof by contrapositive

To prove $P \Rightarrow Q$, instead prove

$$(\sim Q) \Rightarrow (\sim P)$$

by a *direct* proof: assume $\sim Q$, derive $\sim P$.

This is legitimate because of Theorem 1.46 — an implication and its contrapositive are logically equivalent, so proving either proves the other.

You verified that with a truth table in Week 2 and used it as a shortcut in Week 7. Today it becomes a proof technique in its own right.

2. Why you need it

Claim

Let n be an integer. If n^2 is even, then n is even.

Try direct. Assume n^2 is even, so $n^2 = 2k$ for some integer k. Then $n = \sqrt{2k}$ — and now what? You need n to be an integer of the form $2a$, and a square root gives you no such form. The direct route stalls.

Try contrapositive. The contrapositive is: if n is odd, then n^2 is odd.

Proof. Assume n is odd, so $n = 2b + 1$ for some integer b. Then

$$n^2 = (2b+1)^2 = 4b^2 + 4b + 1 = 2\big(2b^2 + 2b\big) + 1.$$

Since $2b^2 + 2b$ is an integer, n^2 is odd. ■

Three lines, and every step is the Wednesday template. **The hypothesis was the problem, not the mathematics** — "n is odd" hands you an equation, and "n^2 is even" does not.

Theorem 3.24

Combining this with the direct proof of the converse gives:

Let $n \in \mathbb{Z}$. Then n^2 is even if and only if n is even.

A biconditional needs two proofs, and here one goes direct and one goes contrapositive. **The two halves need not use the same method.**

3. When to reach for it

The signal

Reach for the contrapositive when the **hypothesis is hard to use** and the **conclusion is easy to negate**.
Typical triggers: the hypothesis involves n^2, n^3, or a product, while the conclusion is a simple parity or sign statement. Negating swaps them, and the easy one becomes your starting point.

Your turn. Prove: if $3n + 11$ is odd, then n is even.

Contrapositive: if n is odd, then $3n + 11$ is even.

Proof. Assume n is odd, so $n = 2k + 1$. Then

$$3n + 11 = 3(2k+1) + 11 = 6k + 14 = 2\big(3k + 7\big).$$

Since $3k + 7$ is an integer, $3n + 11$ is even. ■

4. A proof that proves the wrong thing

Consider this claim and its offered proof.

Claim

Let $n \in \mathbb{Z}$. Then $5n + 7$ is even only if n is odd.

Offered proof — what is wrong with it?

Assume first that n is odd. Then $n = 2a + 1$, so $5n + 7 = 10a + 12 = 2(5a + 6)$, hence $5n + 7$ is even.
Next assume n is even. Then $n = 2b$, so $5n + 7 = 10b + 7 = 2(5b + 3) + 1$, hence $5n + 7$ is odd.

What went wrong. Recall from Week 2 that "P only if Q" means $P \Rightarrow Q$. So the claim is: *if $5n+7$ is even, then n is odd.*

The first paragraph proves the converse. The second proves the contrapositive of the converse — which is the converse again.

Neither paragraph proves the stated claim. The algebra is flawless and the proof is still wrong.

The lesson

Before writing anything, **state precisely which implication you are proving** and what your opening assumption therefore is. Correct algebra attached to the wrong implication earns nothing — and it is much harder to spot than an arithmetic slip.

Exit ticket

1. Write the contrapositive of: "if $7n-3$ is even, then n is odd."

2. In a proof by contrapositive of $P \Rightarrow Q$, what do you assume and what do you derive?

Before Monday

Review Chapters 1, 2, and 5 for the midterm. Monday, October 12 is a review session — bring questions.

Self-check (not collected). Section 3.3, Exercises 1, 3, 5, and 9 — odd-numbered, answers in the back. Exercise 14 is the flawed proof we discussed.

Coming up

Homework 8 is due tonight, Friday, October 9, 11:59 p.m.
Midterm: Wednesday, October 14 — Chapters 1, 2, and 5. **No quiz Monday**; the review session replaces it.
Homework 9 covers Sections 5.5, 3.2, and 3.3, due Friday, October 23.

Questions I still have